

PRACTICAL COURSE WORKBOOK

Building AI Agents & LLM Apps

Build a responsible AI workflow for a real work task

LEVEL	Advanced
GUIDED TIME	5 hours across 12 lessons
PROJECT	a documented prompt, verification and revision workflow
SKILLS	LangChain, AutoGPT, Agents, Critical evaluation, Documentation
EDITION	Structured draft - version 0.9

WHAT YOU WILL BE ABLE TO DO

- Use LangChain appropriately in a realistic, bounded task.
- Use AutoGPT appropriately in a realistic, bounded task.
- Use Agents appropriately in a realistic, bounded task.

WHO THIS IS FOR

People who want dependable, responsible AI-assisted workflows.

BEFORE YOU BEGIN

- Basic computer and web skills

Use this workbook with the online lessons. It is designed for A4 printing, handwritten evidence notes, and offline revision. It does not provide certification.

WORKBOOK GUIDE

A clear route through the course

Building AI Agents & LLM Apps is a structured, practical course covering LangChain, AutoGPT, Agents. It emphasizes explainable methods, checked work and a final outcome that can be reviewed.

MODULE	LESSONS AND FOCUS
01 LangChain: Foundations	1. Capabilities with LangChain 2. Limitations with AutoGPT 3. Responsible use with Agents
02 AutoGPT: Working methods	1. Clear instructions with LangChain 2. Context design with AutoGPT 3. Verification with Agents
03 Agents: Applied workflows	1. Research with LangChain 2. Creation with AutoGPT 3. Automation with Agents
04 LangChain: Quality and review	1. Evaluation with LangChain 2. Privacy with AutoGPT 3. Repeatable practice with Agents

HOW TO USE EACH LESSON PAGE

- 1 Read the objective and identify the decision it supports.
- 2 Attempt the retrieval prompt before checking the online explanation.
- 3 Reproduce the worked example with the stated input.
- 4 Test one normal case and one boundary or failure case.
- 5 Record the result, limitation and next action in the evidence log.

PRINT AND FILE

Print at 100% scale on A4 paper. Use double-sided printing with long-edge binding. Keep completed pages with the project files named in the evidence log.

MODULE 01

LangChain: Foundations

Apply LangChain through capabilities, limitations, responsible use.

LESSON 01 / 18 MIN

Capabilities with LangChain

Learning objectives

- Explain capabilities in the context of Building AI Agents & LLM Apps.
- Apply LangChain to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

capabilities, LangChain, ai-tools

Retrieval prompt

In Building AI Agents & LLM Apps, which evidence best supports a capabilities result produced with LangChain?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 02 / 22 MIN

Limitations with AutoGPT

Learning objectives

- Explain limitations in the context of Building AI Agents & LLM Apps.
- Apply AutoGPT to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

limitations, AutoGPT, ai-tools

Retrieval prompt

In Building AI Agents & LLM Apps, which evidence best supports a limitations result produced with AutoGPT?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 03 / 26 MIN

Responsible use with Agents

Learning objectives

- Explain responsible use in the context of Building AI Agents & LLM Apps.
- Apply Agents to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

responsible use, Agents, ai-tools

Retrieval prompt

In Building AI Agents & LLM Apps, which evidence best supports a responsible use result produced with Agents?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 02

AutoGPT: Working methods

Apply AutoGPT through clear instructions, context design, verification.

LESSON 04 / 30 MIN

Clear instructions with LangChain

Learning objectives

- Explain clear instructions in the context of Building AI Agents & LLM Apps.
- Apply LangChain to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

clear instructions, LangChain, ai-tools

Retrieval prompt

In Building AI Agents & LLM Apps, which evidence best supports a clear instructions result produced with LangChain?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 05 / 18 MIN

Context design with AutoGPT

Learning objectives

- Explain context design in the context of Building AI Agents & LLM Apps.
- Apply AutoGPT to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

context design, AutoGPT, ai-tools

Retrieval prompt

In Building AI Agents & LLM Apps, which evidence best supports a context design result produced with AutoGPT?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 06 / 22 MIN

Verification with Agents

Learning objectives

- Explain verification in the context of Building AI Agents & LLM Apps.
- Apply Agents to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

verification, Agents, ai-tools

Retrieval prompt

In Building AI Agents & LLM Apps, which evidence best supports a verification result produced with Agents?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 03

Agents: Applied workflows

Apply Agents through research, creation, automation.

LESSON 07 / 26 MIN

Research with LangChain

Learning objectives

- Explain research in the context of Building AI Agents & LLM Apps.
- Apply LangChain to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

research, LangChain, ai-tools

Retrieval prompt

In Building AI Agents & LLM Apps, which evidence best supports a research result produced with LangChain?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 08 / 30 MIN

Creation with AutoGPT

Learning objectives

- Explain creation in the context of Building AI Agents & LLM Apps.
- Apply AutoGPT to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

creation, AutoGPT, ai-tools

Retrieval prompt

In Building AI Agents & LLM Apps, which evidence best supports a creation result produced with AutoGPT?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 09 / 18 MIN

Automation with Agents

Learning objectives

- Explain automation in the context of Building AI Agents & LLM Apps.
- Apply Agents to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

automation, Agents, ai-tools

Retrieval prompt

In Building AI Agents & LLM Apps, which evidence best supports a automation result produced with Agents?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 04

LangChain: Quality and review

Apply LangChain through evaluation, privacy, repeatable practice.

LESSON 10 / 22 MIN

Evaluation with LangChain

Learning objectives	- Explain evaluation in the context of Building AI Agents & LLM Apps. - Apply LangChain to a bounded practical task. - Evaluate the result using explicit quality criteria.
Key terms	evaluation, LangChain, ai-tools
Retrieval prompt	In Building AI Agents & LLM Apps, which evidence best supports a evaluation result produced with LangChain?
Evidence to keep	☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 11 / 26 MIN

Privacy with AutoGPT

Learning objectives	- Explain privacy in the context of Building AI Agents & LLM Apps. - Apply AutoGPT to a bounded practical task. - Evaluate the result using explicit quality criteria.
Key terms	privacy, AutoGPT, ai-tools
Retrieval prompt	In Building AI Agents & LLM Apps, which evidence best supports a privacy result produced with AutoGPT?
Evidence to keep	☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 12 / 30 MIN

Repeatable practice with Agents

Learning objectives	- Explain repeatable practice in the context of Building AI Agents & LLM Apps. - Apply Agents to a bounded practical task. - Evaluate the result using explicit quality criteria.
Key terms	repeatable practice, Agents, ai-tools
Retrieval prompt	In Building AI Agents & LLM Apps, which evidence best supports a repeatable practice result produced with Agents?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

CAPSTONE PROJECT

Produce a reviewable Building AI Agents & LLM Apps project using LangChain, AutoGPT, Agents.

DELIVERABLE	A working Building AI Agents & LLM Apps artefact plus an evidence-based self-review.
TOOLS	A suitable LangChain environment, A plain-text decision log, Test data or realistic sample material

Production plan

- 1 Define the intended user, outcome and constraints.
- 2 Create the smallest complete result using LangChain.
- 3 Test one normal case, one boundary case and one failure response.
- 4 Revise the work from the evidence and preserve before-and-after results.
- 5 Prepare a concise handover containing method, limitations and next step.

Definition of done

- The output matches the stated outcome.
- Inputs and decisions are reproducible.
- Boundary and failure evidence is included.
- Limitations and responsibility considerations are explicit.

PROJECT EVIDENCE

Document decisions another person can review

1. PURPOSE AND USER
State the intended user, need, expected outcome and constraints.

2. INPUTS AND ASSUMPTIONS
List source files, data, versions, permissions and assumptions.

PROJECT EVIDENCE

Tests, limitations and revision

3. NORMAL CASE

Expected result, observed result and evidence location.

4. BOUNDARY OR FAILURE CASE

What was difficult, what happened and why it matters.

5. REVISION AND LIMITATION

What changed, what improved and what remains unproven.

REVIEW AND SOURCES

Make the work credible and repeatable

Final self-review

- Can another learner repeat the method?
- Did I test a difficult case?
- Did I avoid unsupported claims?
- Is the next action proportionate to the remaining risk?

FINAL DECISION
Is the project ready to share? State the evidence, remaining risk, owner and next review date.

Authoritative references

AI Risk Management Framework 1.0

NIST - reviewed 2026-08-21
<https://www.nist.gov/publications/artificial-intelligence-risk-management-framework-ai-rmf-10>

AI Principles

OECD - reviewed 2026-08-21
<https://www.oecd.org/en/topics/ai-principles.html>

SOURCE NOTE

Product versions, laws and professional standards can change. Recheck the linked primary source before applying version-sensitive information.

NEXT IMPROVEMENT

Choose one weakness found during review and improve it before extending the LangChain scope.